The Clutchness Hypothesis: A Quantitative Investigation of Repeatability in NBA Late-Game Performance

Prism Research Project

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Abstract

This paper evaluates whether "clutch" performance is a repeatable player skill or largely statistical noise amplified by narrative. Using eight NBA seasons (2017–18 through 2024–25), clutch performance is defined as possessions in the last five minutes with score differential ≤ 5 points. We model repeatability with year-over-year Pearson correlation of clutch true shooting percentage (TS%), and we run complementary tests on sample size, shot quality proxies, free-throw dependence, usage concentration, turnover behavior, and home/away effects. The central estimate is weak repeatability ($r = 0.074$ regular season; $r = 0.15$ playoffs), with small practical effect despite statistical detectability in the large regular-season sample. Across 22 reputation players, 19 perform worse in clutch than overall efficiency, while 102 non-reputation players outperform baseline efficiency in clutch contexts. Results support the view that "clutchness" is dominated by low-volume variance, role concentration, and selection effects rather than stable shooting skill.

1 Research Question and Hypotheses

Research question: Is clutch scoring efficiency a persistent individual skill from one season to the next?

Null hypothesis (H_0): Clutch TS% shows no meaningful year-over-year consistency, $|r| < 0.3$.

Alternative hypothesis (H_1): Clutch TS% shows meaningful year-over-year consistency, $|r| \geq 0.3$.

2 Data and Operational Definition

Source endpoints: NBA Stats API `LeagueDashPlayerClutch` and `LeagueDashPlayerStats`.

Window: 2017–18 through 2024–25 (8 seasons).

Clutch definition:

- Last 5 minutes of regulation/overtime
- Score margin within 5 points (ahead or behind)
- Regular season and playoffs analyzed separately where available

Inclusion rules:

- Regular-season baseline filter: players with at least 15 clutch games
- Playoff baseline filter: players with at least 5 clutch games
- Reputation subset: 22 players with widely cited clutch narratives

3 Statistical Modeling Framework

3.1 Core Efficiency Metrics

True Shooting Percentage (TS%):

$$\text{TS\%} = \frac{\text{PTS}}{2(\text{FGA} + 0.44\,\text{FTA})}$$

Approximate possessions:

$$\text{Poss} = \text{FGA} + 0.44\,\text{FTA} + \text{TOV}$$

FT-stripped efficiency:

$$\text{FT-Stripped} = \frac{2(\text{FGM} - \text{FG3M}) + 3\text{FG3M}}{2\text{FGA}}$$

3.2 Repeatability Model

Let x_i be clutch TS% in season N and y_i be clutch TS% in season $N+1$ for the same player-season pair.

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}$$

Interpretation thresholds used in this project:

- $|r| < 0.3$: weak repeatability
- $0.3 \leq |r| < 0.5$: moderate repeatability
- $|r| \geq 0.5$: strong repeatability

3.3 Ancillary Behavioral Models

To probe mechanism rather than just outcomes:

- Usage concentration: FGA Share $= \frac{\text{Player FGA}}{\text{Team FGA}}$
- Ball movement proxy: AST/FGM
- Turnover rate: $\frac{\text{TOV}}{\text{FGA}+0.44\,\text{FTA}+\text{TOV}}$
- Shot selection proxy: 3PA Rate $= \frac{\text{FG3A}}{\text{FGA}}$

4 Results

4.1 Primary Repeatability Test

Table 1: Year-over-year clutch TS% repeatability

Metric	Regular Season	Playoffs
Pearson r	0.074	0.15
p-value	0.005	0.32
Adjacent-season pairs	1,435	46
Interpretation	Weak	Weak

The regular-season estimate is statistically detectable but practically small; in practical forecasting terms, clutch TS% in season N has minimal predictive power for season $N + 1$.

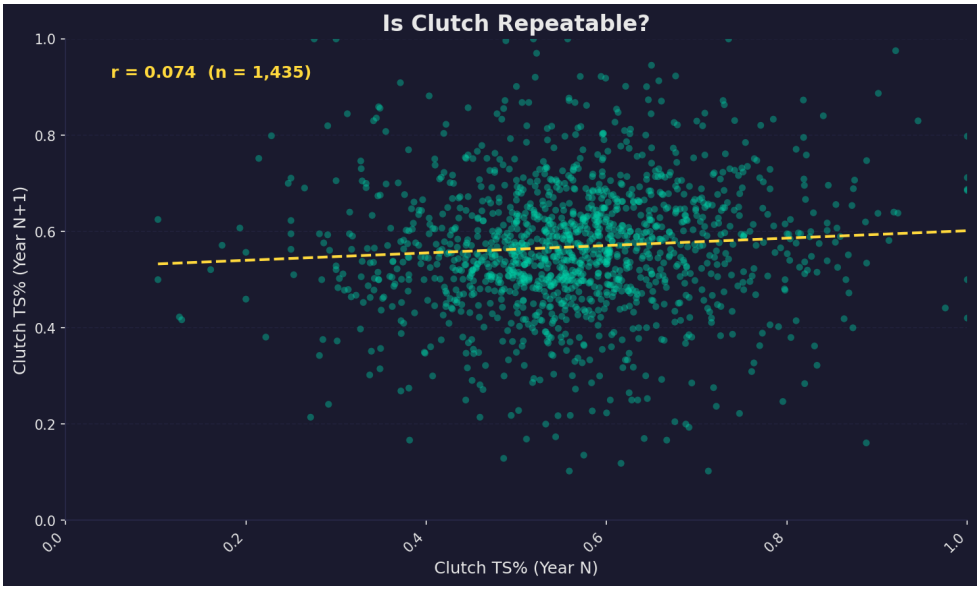

Figure 1: Year N vs Year $N+1$ clutch TS%. The near-flat trend and diffuse cloud indicate weak repeatability.

4.2 Sample-Size Constraint (Signal-to-Noise)

Table 2: Clutch volume relative to total offensive workload

Metric	Regular Season	Playoffs
Average clutch possessions / player-season	34.5	11.9
Median clutch possessions	25.1	—
Average overall possessions	817.2	—
Clutch / overall possession ratio	3.7%	4.8%
Average clutch games played	23.9	7.3

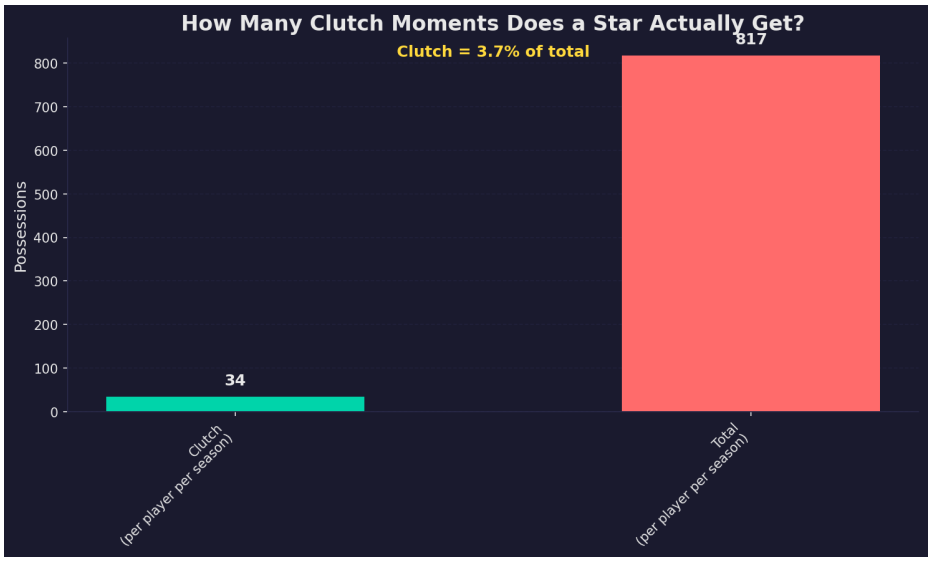

Figure 2: Clutch possessions are a small fraction of total possessions, limiting precision of individual estimates.

4.3 Reputation-Player Stress Tests

Across 22 reputation players, 19 post lower clutch TS% than overall TS%. This undermines a broad "stars get better in clutch" claim.

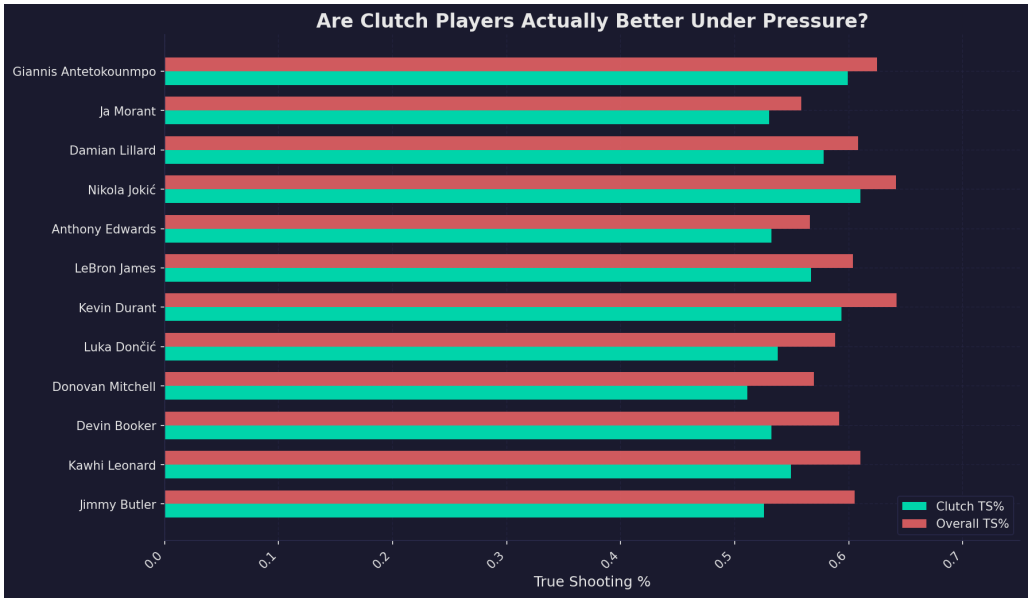

Figure 3: Clutch vs overall TS% for reputation players. Most clutch values are below season baseline.

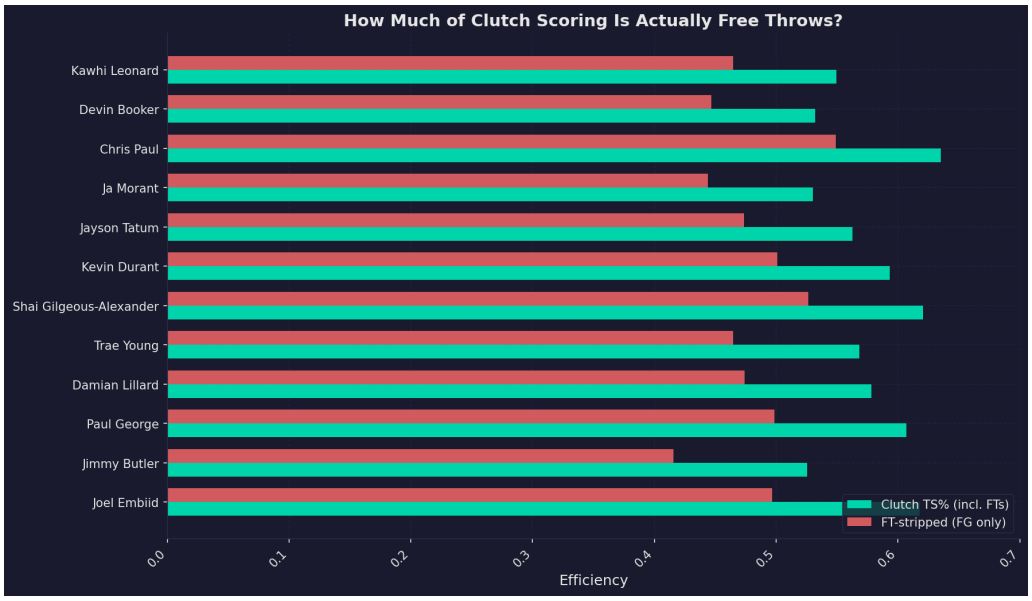

Figure 4: Clutch TS% vs FT-stripped efficiency. Gaps suggest free-throw dependence in some clutch narratives.

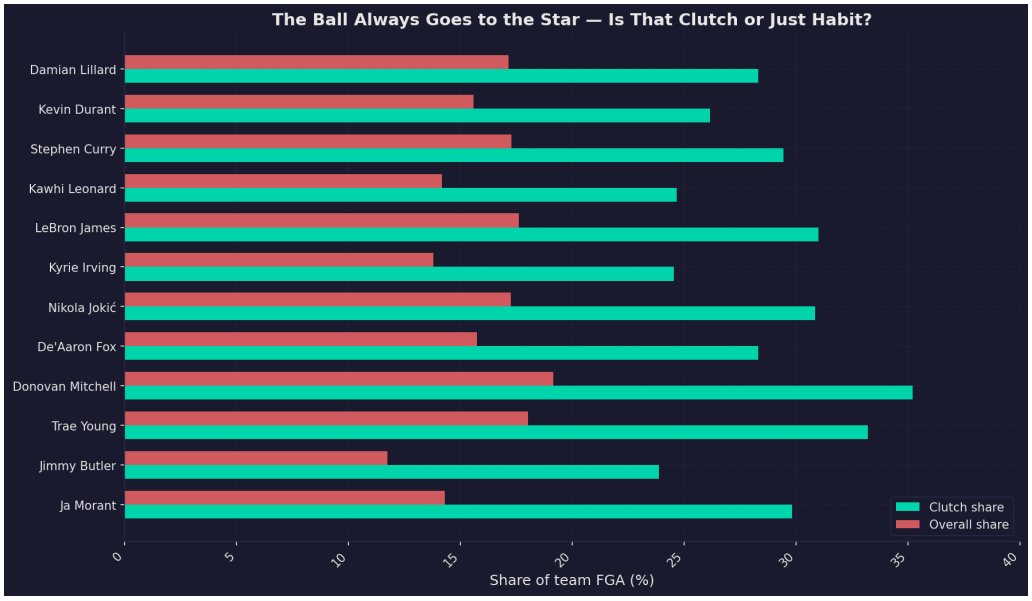

Figure 5: Usage share spikes in clutch time, indicating role concentration.

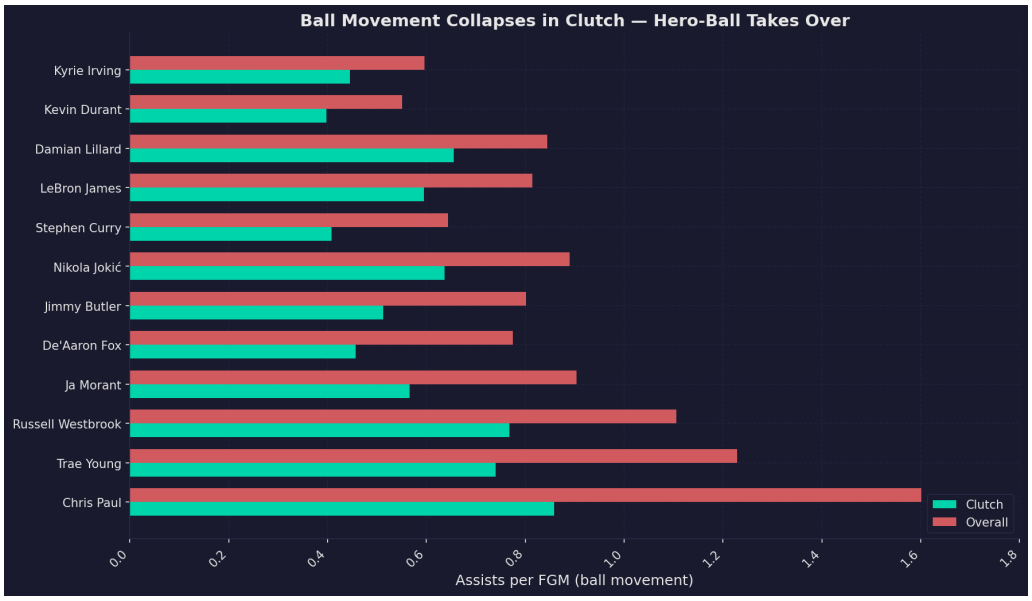

Figure 6: Assist-to-FGM ratio generally falls in clutch contexts, consistent with increased isolation possessions.

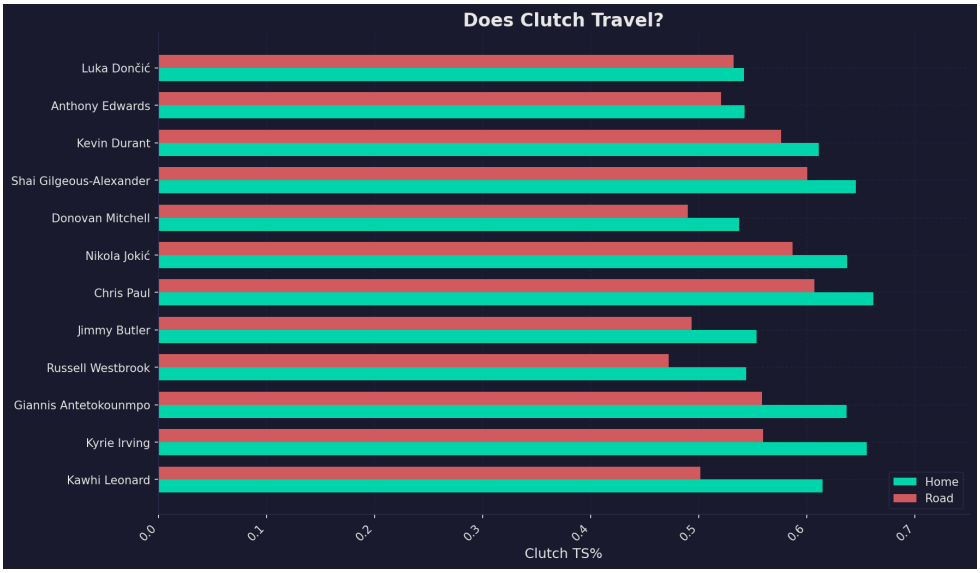

Figure 7: Home/away clutch TS% effects are mixed across players, not directionally uniform.

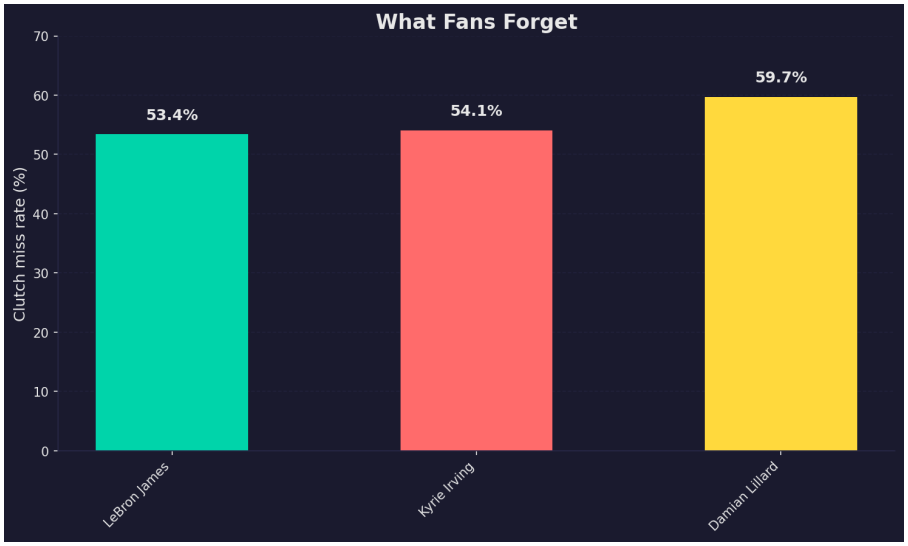

Figure 8: Miss-rate view of high-profile players: memorable makes coexist with frequent misses.

4.4 Hidden Clutch and Counter-Narrative Evidence

The analysis identifies 102 non-reputation players with better clutch than overall efficiency, and 17 non-stars with strong repeatability ($r > 0.5$), suggesting that narrative attention and media salience are highly selective.

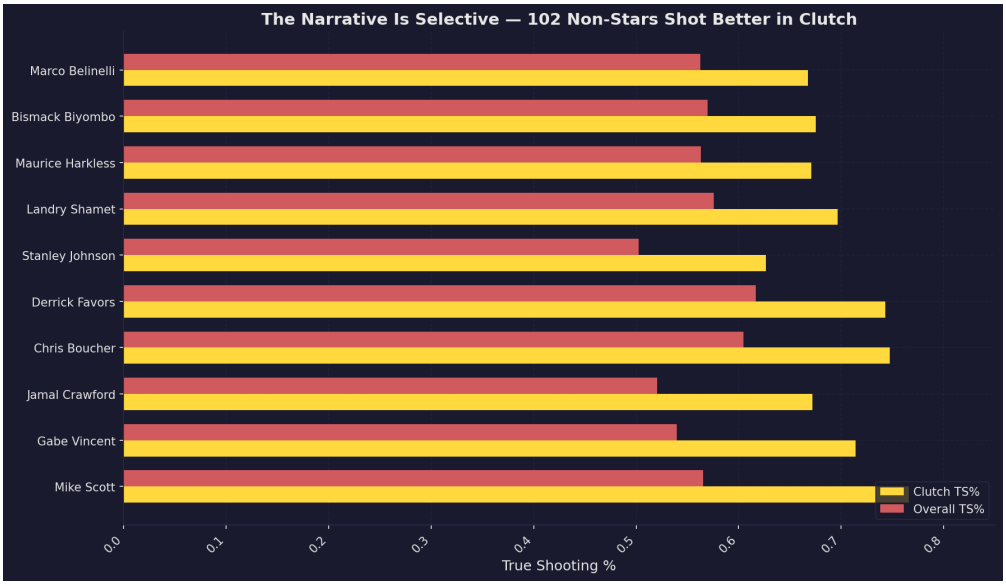

Figure 9: Top non-reputation players outperforming baseline efficiency in clutch contexts.

4.5 Turnovers and Decision Quality

Nineteen of 22 stars show lower turnover rates in clutch than overall. This implies the clutch TS% decline is not primarily driven by panic errors; it is more consistent with reduced shot quality and defensive load.

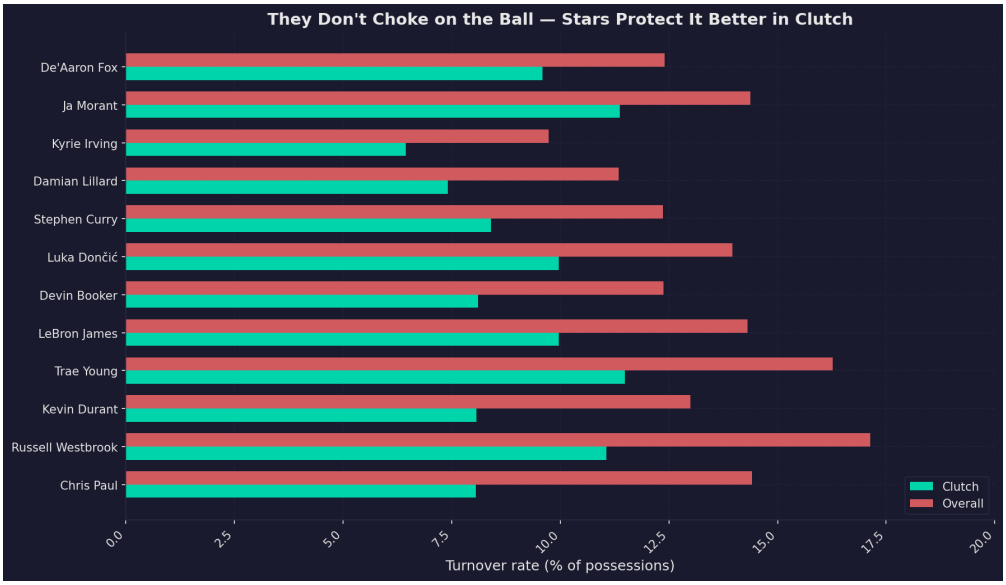

Figure 10: Clutch vs overall turnover rates among reputation players.

4.6 Notable Exception

Chris Paul appears as a clear positive outlier with a sizeable clutch edge (+5.2 percentage points TS%) and sharply improved clutch ball security.

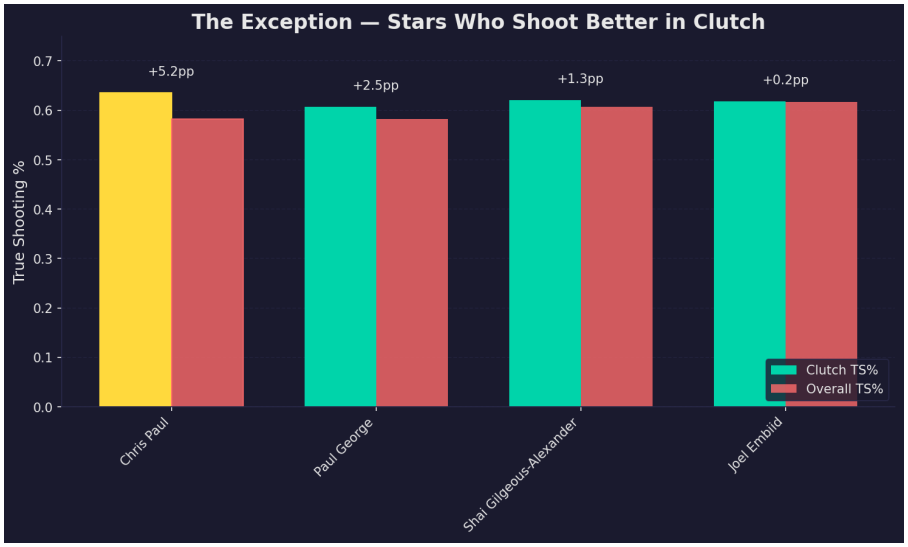

Figure 11: Three players with clutch TS% above overall TS%; Chris Paul is the strongest exception.

5 Interpretation and Implications

Main inference: Clutch shooting performance is weakly repeatable at the player-season level under this definition.

Mechanistic interpretation:

- *Low-volume regime:* clutch possessions are too sparse for stable individual estimation.
- *Usage concentration:* stars absorb higher-volume, higher-attention shots in clutch moments.
- *Style shift:* reduced assist ratios indicate more self-created attempts against set defenses.
- *Narrative asymmetry:* high-salience makes are remembered disproportionately versus routine misses.

6 Limitations

- Clutch definition (last 5 minutes, within 5) is standard but still arbitrary.
- Playoff sample is small (46 adjacent-season pairs), limiting power.
- Potential confounders (defensive scheme, lineup quality, fatigue, era/rules) are not fully controlled.
- Selection dynamics may remove poor performers from future clutch opportunities (survivorship bias).

7 Conclusion

Clutch *moments* are real and high leverage, but clutch as a stable, repeatable shooting skill is weakly supported by these data. The quantitative evidence points to variance, context, and role structure as the dominant drivers of observed clutch outcomes.

Reproducibility Assets

Key project artifacts are available in the workspace:

- `Clutchness Hypothesis/outputs/phase2_results.json`
  `Clutchness Hypothesis/outputs/phase3_results.json`
- `Clutchness Hypothesis/outputs/phase2_yoy_pairs.csv`
- `Clutchness Hypothesis/outputs/phase3_test3_ts_comparison.csv`
- `Clutchness Hypothesis/outputs/phase3_test4_ft_stripped.csv`
- `Clutchness Hypothesis/outputs/phase3_test5_usage_spike.csv`
- `Clutchness Hypothesis/outputs/phase4_test6_assist_ratio.csv`
- `Clutchness Hypothesis/outputs/phase4_test7_home_away.csv`
- `Clutchness Hypothesis/outputs/phase4_test8_miss_rate.csv`
- `Clutchness Hypothesis/outputs/turnover_rate_clutch.csv`